

Quantitative determination of lignans and isoflavonoids in plasma of omnivorous and vegetarian women by isotope dilution gas chromatography-mass sp...

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The first quantitative method for the determination of both lignans and isoflavonoid phytoestrogens in plasma is presented. Using ion-exchange chromatography the diphenols are separated into two fractions 1) the biologically "active" fraction containing the free compounds + mono- and disulfates and 2) the biologically "inactive" fraction containing the mono- and diglucuronides and the sulfoglucuronides. After hydrolysis the fractions are further purified by solid phase extraction and ion exchange chromatography. Losses during the complete procedure are corrected for using radioactive estrogen conjugates during the first steps and later by adding deuterated internal standards of all compounds measured (matairesinol, enterodiol, enterolactone, daidzein, O-desmethylangolensin, equol, and genistein). The final determination is carried out by isotope dilution gas chromatography-mass spectrometry in the selected ion monitoring mode (GC/MS/SIM). The diphenols may be measured at concentrations as low as 0.2 to 1.0 nmol/l. Results of plasma analyses of all compounds in 27 pre- and postmenopausal omnivorous and vegetarian women are presented for the first time. The most important findings are that the free+sulfate fraction is low for genistein (3.8% of total), but as much as 21-25% of enterolactone and enterodiol occurs in this fraction. A good correlation between plasma and urine values was found. Total concentrations of individual compounds vary greatly between the subjects (from pmol/l to μ mol/l), the vegetarians having higher values, particularly one vegan subject. The highest total enterolactone concentration value exceeded 1 μ mol/l. It is concluded that a highly specific method for the assay of 3 lignans and 4 isoflavonoids in plasma has been developed. This method will be useful in future studies of lignan and isoflavonoid metabolism.